Inverse Earth Studio: a reproducible geophysical workbench for processing, forward modelling, inversion, and scientific review

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Abstract

Geophysical inversion is underdetermined: observations constrain a subsurface model only through acquisition geometry, a forward operator, error assumptions, parameterization, prior information, and numerical optimization. Inverse Earth Studio is an open-source research workbench connecting data ingestion and processing, forward modelling, inversion, diagnostics, and interpretation across potential fields, electromagnetics, electrical methods, and seismology. Its implementation includes Python pipelines and solvers, bilingual computational notebooks, provenance-bearing experiments, browser instruments, and a bounded authenticated-service design.

The method suite spans gravity and magnetic processing and inversion, magnetotelluric screening and layered-earth inversion, electrical-resistivity quality control and inversion, waveform processing and phase picking, travel-time inversion, acoustic full-waveform inversion, joint inversion, and learned estimators. Results include both recoveries and failures. For example, a synthetic gravity case has weighted WRMS 1 while recovered model correlation is 0.283 and centroid depth is about 570 m for a 282 m truth; a low-velocity FWI case remains unresolved. Clear Lake station cl061 is retained for tensor quality control and excluded from 1D inversion after failing the consistency screen. The current BS-1 travel-time refinement study is a negative synthetic result: on the joint family/acquisition holdout, two same-seed CUDA runs have velocity RMSE ratios of 2.194 and 2.190, and independent travel-time RMSE ratios of 5.585 and 5.595 relative to the selected weighted-L2 reference. Every joint-shift case exceeds its run-specific validation threshold. These are fixed synthetic cohorts, not field-performance estimates or independent-training-seed evidence. Null-space projection is treated as established prior methodology, not as a novel contribution of this product.

The workbench makes data fit, model recovery, and geological interpretation separate questions. Synthetic cases carry known truth; field observations do not. Replayed browser artifacts are distinguished from online calculations, and notebook execution is distinguished from scientific validation. The report documents the methods, numerical assumptions, data provenance, executable results, and limitations; it does not claim that unresolved scientific or service-acceptance gates have passed.

1 Scientific motivation

An inverse image can look geologically persuasive even when multiple models explain the data, the assumed physics are violated, or a withheld acquisition subset is poorly predicted. This is familiar across geophysics: potential-field inverses are non-unique; MT dimensionality and component quality condition whether a 1D interpretation is defensible; seismic inversion depends on illumination, bandwidth, starting model, and cycle skipping; and coupled or learned estimates inherit their structural or training priors. A low objective is evidence about a specified objective under a specified data model. By itself, it is not evidence of unique or accurate geology.

The topic sequence draws on the public theoretical-practical geophysical inversion course by Mantilla. Its notebooks, recordings, images, and raw data are not redistributed. The source commit was `0084021fcc6c1d2f112b144e6e5f0a3d92f20dff` (Mantilla course). The research survey distinguishes primary sources from implementation choices and identifies where the methods differ from cited work (docs/research/review.md). The aim is to connect domain assumptions, executable calculations, diagnostic evidence, and failure states within one inspectable project.

The report covers gravity station correction and prism inversion, aeromagnetic processing, induced-magnetization modelling, magnetotelluric screening and layered-earth inversion, electrical resistivity inversion, response-aware waveform processing, first-arrival tomography, acoustic full-waveform inversion, joint potential-field inversion, learned seismic-velocity estimation, and earthquake phase picking. It also evaluates a proposed coverage-conditioned null-space correction. Method definitions and numerical assumptions are in docs/design/SDD.md.

2 Software and evidence design

2.1 Evidence chain

The intended data flow follows the path

`source → immutable raw asset → validated dataset → processing → model/survey → solve → evaluation → result/export`.

At each boundary the contracts carry modality, named dimensions and order, coordinate frame or specified local coordinates, units, geometry, masks and reasons, uncertainty, parent identities, code/environment information, and hashes as applicable. Unknown units, datum, signs, component orientation, rights, or response conventions are reasons to reject or quarantine an input rather than infer missing facts. Known truth is attached only to explicitly synthetic cases. Field observations may be screened and processed, but a synthetic constructor is never substituted for missing field provenance or truth.

The application separates three execution contexts. Visitors can inspect public material and use only those browser calculations whose client-side behavior has been validated. Authenticated project workflows are designed to isolate upload, ownership, and bounded worker execution on the single canonical ML VPS origin. Larger training, extensive FWI, and full bake workflows remain local CPU/GPU pipelines. A static or browser replay of a saved artifact is not represented as an online inverse solve. The specified deployment uses one bounded ML VPS service. Its capacity and numerical job behavior have not been measured on the live host.

2.2 Reproducible software artifacts

The software contains a Python data pipeline and solver implementations, generated and source-controlled scientific outputs, JSON contracts and validation outputs, frontend TypeScript/React instruments, and computational notebooks in English and Spanish. The user-facing browser is intended to display computed arrays and explicit physical axes, not to invent a second approximate result for animation. Physical result records distinguish observations, predictions, signed residuals, model values, objective terms, coverage and uncertainty diagnostics, stopping state, and provenance. Display interpolation or color mapping is a rendering operation and must not be mistaken for increased model resolution.

Notebooks are part of the research material, not merely screenshots. Their lessons explain assumptions, execute the named calculations, show numerical outputs, compare with independent identities or baselines when available, and retain unresolved results. The bilingual content was corrected where translations, counts, or numerical interpretations differed. Execution and translation checks establish neither field validity nor educational efficacy.

3 Data and experimental scope

3.1 Authored synthetic experiments

The original canonical suite uses independently constructed geological geometries instead of repeating one centered anomaly. Its earlier 0.04 report describes 20 distinct cases and six controlled conditions per case (120 experiments, 348 method results). The conditions vary contrast, noise, acquisition, coverage, or regularization according to the physical family; they are not interchangeable perturbations and are not samples from a geological population. Later repository work adds separate survey, uploaded-data, learning, and source-screening experiments. Therefore the older 120-experiment suite is a defined artifact family, not a count of every experiment now present in the repository.

Synthetic constructors provide known truth for controlled tests of forward physics and conditional inversion behavior. They do not encode the full complexity of field geology. Noise perturbations at fixed mesh/prior/tradeoff measure conditional estimator repeatability, not posterior uncertainty. A held-out line, shot, family, or acquisition tests only the specified split and comparator. Repeated numerical trials from one generator do not establish field performance.

3.2 External data and eligibility

Two SimPEG tutorial archives are independently pinned and locally preprocessed as ingestion examples. The source documentation states that their derived observation tables are not redistributed while upstream redistribution rights remain unresolved. They are not field validation. The measured Clear Lake EDI station cl061 is hash-pinned and cited to the USGS and EarthScope data releases [15, 16]. Its impedance tensor fails the required 1D consistency screen; it remains a QC-only dataset, and no layered inversion or geological result is claimed.

Other data paths include provider-linked or rights-restricted data. For example, the Slagdump ERT raw asset is absent locally under its provider-link-only status; focused notebook checks consequently skip tests that require those bytes. The Koenigsee travel-time lesson

is an authored synthetic method control and does not establish execution on the provider-linked case. The M13 STEAD excerpts [6] and associated project-trained checkpoint have a scoped redistribution decision with attribution; that decision does not generalize to other data or establish model accuracy. Rights, data custody, field eligibility, numerical correctness, and scientific validation remain separate decisions.

4 Methods

The method identifiers below follow docs/design/SDD.md. Each entry names an implementation and its tests; this does not imply validated use for every input or a complete end-to-end workflow.

Method	Executable purpose	Principal limitations
Gravity corrections and transforms	Gravity station corrections, datum and uncertainty lineage, and transforms with safeguards against duplicate corrections.	Field eligibility still depends on exact source, datum, instrument, and uncertainty lineage. A local adapter pass does not validate arbitrary field tables.
Weighted 3D gravity prism inversion	Weighted 3D gravity prism forward/inverse calculations, including a bounded L2 survey inversion and comparison with independent prism physics.	Non-uniqueness persists; a good data fit or a visually compact body does not identify geology. Full-survey and field performance remain untested.
Aeromagnetic processing and continuation	Aeromagnetic line/tie-line/height checks, blocked line holdout, support-aware gridding, and physically supported upward continuation.	Authored survey controls do not establish broad provider-data validation or field magnetic inversion.
Induced-magnetization prism modelling	Induced-magnetization prism forward modelling and physical objective/derivative construction, checked against Choclo.	A full nonzero survey inverse with held-out evaluation is not included in the available evidence. Remanence violates the induced-direction assumption.
EDI quality and dimensionality screening	MT EDI component, variance, rotation, tipper, and dimensionality screening.	A tensor record is not automatically eligible for a 1D inverse. cl061 remains QC-only.
Layered-earth impedance inversion	Layered-earth complex-impedance response and bounded 1D inversion, with cross-checks against MTpy and browser/local analytic cases.	Known layer thickness and isotropic 1D assumptions constrain the synthetic inverse. The cl061 field screen fails.
Two-dimensional electrical resistivity inversion	2D ERT processing/inversion with electrode geometry, topography, reciprocal/QC logic, blocked holdout, and explicit failed-inverse states.	Rights-restricted raw data are not fabricated or represented as executed; independent comparison and some field paths remain open.
Response-aware waveform processing	MiniSEED/StationXML waveform handling, response-aware filtering, STA/LTA or matched-arrival picks.	Response epochs, units, gaps, and channel orientation are strict eligibility conditions. STA/LTA picks are not universal truth labels.
First-arrival travel-time inversion	First-arrival travel-time inversion with explicit source/receiver geometry, bounded velocity, ray coverage, and held-out-shot comparison against a baseline.	The executed teaching control is synthetic and uses a straight-ray/Dijkstra model; the synthetic control does not represent provider-linked field data.

Method	Executable purpose	Principal limitations
Acoustic full-waveform inversion	Acoustic constant-density FWI with Deepwave, bounded velocity updates, full-band and frequency-continuation schedules, active/withheld receiver diagnostics, and saved optimizer states.	Non-convexity, cycle skipping, limited illumination, inverse crime from using the same forward engine, and acoustic-only physics remain material limitations. Some cases improve model RMSE but fail the specified noise-consistency criterion.
Joint potential-field inversion	Joint potential-field inversion comparing uncoupled fits, spatial cross-gradient coupling, and an explicit Gaussian-mixture petrophysical prior.	The project Gaussian-mixture-prior implementation is not the SimPEG PGI optimizer. Coupling can trade one property against another; one calibration matrix does not establish general performance.
Learned seismic-velocity estimation	A learned seismic-velocity inversion trained on disjoint procedural families and compared with a same-input weighted L2 solve and independent quadrature forward operator.	The frozen joint family/acquisition holdout fails both prespecified ratios; training success does not override the held-out result.
PhaseNet-family phase picking	PhaseNet-family P/S inference [7], matched STA/LTA comparison, and local browser ONNX inference with explicit preprocessing and abstention rules.	Curated browser parity is runtime consistency, not accuracy. Population metrics come from the separate locked test cohort and analyst references, not geological truth.
Geometry-normalized travel-time residual refinement	Backprojected travel-time residuals are normalized by ray length and cell coverage, then supplied with the coverage field to a bounded residual CNN.	The current physics-v2 study fails both predeclared joint-shift ratios in the primary and same-seed replay. This is a negative synthetic comparison; null-space projection is established prior methodology and no superiority claim is made.

4.1 Method-specific mathematical formulations

Gravity station corrections and uncertainty

The station workflow treats a gravity value as a measurement with a history, not as an unlabeled scalar. Each row retains the source value, station identifier, coordinate reference, elevation convention, unit, instrument or provider correction state, and uncertainty. A correction is applied only when the source record and selected operation establish that it has not already been applied. Unknown metadata are preserved as unknown and cause the corresponding derived quantity to be withheld; the code does not infer a datum or silently convert a likely unit. This fail-closed rule matters because a numerically plausible anomaly can be produced from an incorrect sign, a duplicated free-air term, a mixed reference datum, or a coordinate error.

For a station record i, the processed anomaly can be represented as the source observation plus a vector of signed corrections and a reference shift,

$$g_i^{\mathrm{proc}} = g_i^{\mathrm{src}} + \mathbf{c}_i^{\mathsf{T}}\mathbf{a} + \Delta g_{\mathrm{datum}}, \tag{1}$$

where $\mathbf{c}_i$ contains only corrections supported by the declared input and $\mathbf{a}$ contains their sign convention. This notation is an accounting identity, not an assertion that every source provides every correction. A missing correction value is not equivalent to zero. The implementation keeps the source and derived columns separate so a correction chain can be reproduced and audited. The no-double-correction validator rejects a record whose declared state is inconsistent with the requested chain before the output is emitted.

If source and correction errors are independent, the propagated variance is the sum of their contributions. More generally, the declared covariance is transformed by the same linear operation,

$$\sigma^2_{i,\mathrm{proc}} = \mathbf{a}_i^{\mathsf{T}}\,\mathbf{C}_i\,\mathbf{a}_i, \qquad \mathbf{C}_{\mathrm{proc}} = \mathbf{A}\,\mathbf{C}_{\mathrm{src}}\,\mathbf{A}^{\mathsf{T}}. \tag{2}$$

Here $\mathbf{a}_i$ is the local coefficient vector for the scalar record, whereas $\mathbf{A}$ is the complete linear map over a stacked survey. A diagonal-only uncertainty output is justified only when the input specifies independent errors or when the transform preserves that assumption. The processor does not manufacture cross-station covariance from marginal standard deviations.

Implementation is split between source parsing, unit and datum checks, correction-state validation, deterministic calculation, and a result receipt. The receipt binds each output to its parent asset and records the applied correction sequence, uncertainty provenance, and the implementation identity. The station tests exercise a known correction lineage, unit conversion, reference handling, uncertainty propagation, and rejection of duplicated or unknown operations. These tests prove the declared cases and code paths; they do not establish that an arbitrary downloaded table is a complete or eligible gravity survey.

The lesson is deliberately connected to the potential-field inverse that follows. A station table can pass syntax checks while remaining scientifically inadmissible because of an undocumented tide convention, unresolved datum, missing terrain model, or provider-level correction that cannot be reconstructed. The interface therefore separates raw records from processed records and exposes the provenance of each column. A user must be able to determine which values came from the provider, which were calculated locally, and which remain unavailable. No corrected anomaly is presented as subsurface density, and no correction step is described as an inversion. The relevant executable entry point is data-pipeline/gravity_processing.py; the stable station contract and rejection tests are in tests/numerics/test_gravity_processing.py.

Weighted three-dimensional gravity prism inversion

The gravity inverse maps a discretized density contrast to observations through a forward operator assembled from three-dimensional prism responses. Let $\mathbf{m}$ contain cell density contrasts and let $\mathbf{G}$ map those contrasts to vertical gravity at the observation locations. For the declared linear density model, the prediction is

$$\mathbf{d}_{\mathrm{pred}} = \mathbf{G}\mathbf{m}, \qquad \mathbf{r} = \mathbf{d}_{\mathrm{obs}} - \mathbf{G}\mathbf{m}. \tag{3}$$

The units of $\mathbf{m}$, geometry, coordinates and $\mathbf{d}$ are checked at the interface. The mesh order is part of the numerical contract; a vector reshaped in a different cell order may look smooth but represents different physical cells.

The bounded L2 reference minimizes a noise-weighted data term together with a model-space penalty,

$$\Phi(\mathbf{m}) = \tfrac{1}{2}\|\mathbf{W}_d(\mathbf{G}\mathbf{m} - \mathbf{d}_{\mathrm{obs}})\|_2^2 + \tfrac{\beta}{2}\|\mathbf{W}_m(\mathbf{m} - \mathbf{m}_{\mathrm{ref}})\|_2^2, \qquad \mathbf{m}_{\min} \le \mathbf{m} \le \mathbf{m}_{\max}. \tag{4}$$

The reference model, bounds, weighting, regularization, mesh, active cells, and stopping conditions are reported as parameters of a run. Weights may encode stated standard deviations, but they do not transform a deterministic synthetic perturbation into a probability model. The minimizer is conditional on the mesh, reference and penalty. The reported model is not a unique geological image.

For a linear forward model the gradient and normal equations provide an independently checkable oracle,

$$\nabla\Phi(\mathbf{m}) = \mathbf{G}^{\mathsf{T}}\mathbf{W}_d^{\mathsf{T}}\mathbf{W}_d(\mathbf{G}\mathbf{m} - \mathbf{d}_{\text{obs}}) + \beta\mathbf{W}_m^{\mathsf{T}}\mathbf{W}_m(\mathbf{m} - \mathbf{m}_{\text{ref}}). \tag{5}$$

The project checks selected forward values against an independent prism calculation, then compares the SimPEG-backed solve with a bounded least-squares reference on the same data, survey, mesh and error model. This separates a forward-kernel error from optimizer configuration and from differences caused by changing the data or discretization.

Evaluation reports fitted and held-out predictions, weighted residual statistics, sensitivity or coverage diagnostics, and model-recovery metrics only when synthetic truth exists. The reference case demonstrates why these quantities are not interchangeable: a weighted WRMS near one coexists with a model correlation of 0.283 and a recovered centroid near 570 m for a truth near 282 m. The datum fit is consistent with the chosen scale, while the density distribution is unresolved. Potential-field depth/volume tradeoffs and mesh dependence remain; a single model slice cannot remove them. The implementation is data-pipeline/gravity_l2.py and the bounded calibration is covered by tests/numerics/test_gravity_l2.py and tests/numerics/test_gravity_l2_selection.py. The full family of solver outcomes, including ineligible or unresolved records, must be retained in the result catalogue rather than reduced to its best-looking case.

Aeromagnetic processing and harmonic continuation

The aeromagnetic workflow begins with flight-line geometry and provider leveling provenance. It checks coordinates, line identifiers, sample spacing, flight height, tie-line support, duplicate points, and data gaps before interpolation. A blocked flight line is reserved for evaluation before gridding; the withheld observations are not used to choose a grid or tune its smoothing. This test is intentionally different from randomly withholding nearby points, since spatially adjacent samples can share the same acquisition error and geological signal.

Potential fields in a source-free half-space satisfy Laplace's equation. For a horizontal Fourier component with wavenumber magnitude k, upward continuation by height h applies the transfer function

$$\widehat{T}(k_x, k_y, h) = \widehat{T}(k_x, k_y, 0)\exp(-h\sqrt{k_x^2 + k_y^2}). \tag{6}$$

Thus higher wavenumbers decay faster. The operation suppresses short wavelength content and is not an inverse continuation: recovering the surface field from an elevated grid would amplify noise. The implementation validates the transfer against a single analytic harmonic with known wavenumber and continuation height; the test is a physical identity, not an assertion that a gridded field uniquely determines its sources.

Interpolation and support are represented separately. Let $\mathcal{S}$ be the sampled footprint and $\mathcal{D}$ the output grid. A prediction outside an accepted support neighborhood is stored as unavailable, not silently filled with a value that looks measured. The gridding receipt records the interpolation rule, node spacing, coordinate frame, input line selections, mask, and any height transform. This permits a reviewer to distinguish the actual flight path

from the raster display. The interface displays both the raster and the original lines, along with the blocked-line residuals and support mask.

The current public USGS Amargosa example uses the provider-levelled channel without repeating upstream diurnal, IGRF, navigation, or flight/tie-line levelling corrections. Its rights and source hash are recorded in the research evidence; the calculation is a bounded processing/QC example, not independent site validation or a magnetic inversion. Upstream corrections cannot be reconstructed from absent instrument and processing records. A different provider dataset would need its own line geometry, units, leveling history, datum, and rights decision before being processed.

The source implementation separates parsing, line selection, blocked holdout, support-aware gridding, continuation, and output provenance. The core processing functions are in data-pipeline/magnetic_processing.py, while upward continuation and its independent transfer check are in data-pipeline/magnetic_continuation.py and tests/numerics/test_magnetic_continuation.py. The measurable acceptance evidence is the held-out line and analytic harmonic; a smooth map alone is not a processing validation. Flight-line texture, mask boundaries, physical units and the acquisition track should remain visible in scientific views so that interpolated values cannot be mistaken for direct observations.

Induced-magnetization prism modelling

The induced-field model assumes that each active cell's magnetization is parallel to a declared inducing field. For cell susceptibilities $\boldsymbol{\chi}$ and a fixed field direction $\hat{\mathbf{h}}_0$, the cell magnetization is proportional to $\chi_j \mathbf{H}_0$ under the stated linear, isotropic assumption. A prism kernel maps each cell to the vector magnetic field at receiver i; the scalar total-field response is a first-order projection onto the inducing direction,

$$\Delta T_i \simeq \hat{\mathbf{h}}_0^{\mathsf{T}} \Delta \mathbf{B}_i, \qquad \Delta \mathbf{B}_i = \sum_j \mathbf{K}_{ij}(\mathcal{G})\, \chi_j \mathbf{H}_0. \tag{7}$$

Here $\mathcal{G}$ includes prism bounds, coordinates, and the local frame. The approximation is the total-field linearization appropriate to the declared weak-anomaly model; the project does not reinterpret it as a recovered vector magnetization. Remanent magnetization that is not parallel to the inducing field violates the assumption and is a negative-control case.

The fit objective is assembled from the physical forward predictions and a covariance model for the observation errors. For residual $\mathbf{r}(\mathbf{m}) = \mathbf{d}_{\text{pred}}(\mathbf{m}) - \mathbf{d}$,

$$\Phi(\mathbf{m}) = \tfrac{1}{2}\mathbf{r}^{\mathsf{T}}\mathbf{C}_d^{-1}\mathbf{r} + \beta\, \Phi_m(\mathbf{m}), \qquad \nabla\Phi = \mathbf{J}^{\mathsf{T}}\mathbf{C}_d^{-1}\mathbf{r} + \beta \nabla\Phi_m. \tag{8}$$

The covariance may contain correlated errors; replacing it by a diagonal weight matrix changes the likelihood and therefore the inverse problem. The executable assembly binds the survey, sensitivity, covariance, model parameterization, prior, source implementation and selected likelihood into one objective identity. Derivative tests compare the analytical gradient with finite differences and a separate Choclo field calculation on fixed geometries.

The currently retained 528-cell linear-TMI development fold has a source-bound objective and oracle certificate. It is a bounded synthetic fold, not completed survey inversion: candidate selection, sparse and nonlinear solves, held-out evaluation, field qualification, final export, and online resource admission remain unresolved. A converged fold would establish only that a specified numerical solve met its conditions; it would not validate remanence, geology, the inducing-field record, or performance on a real survey.

The front end and report must expose the receiver geometry, inducing field vector and coordinate convention, model units, active-cell mask, predicted total-field anomaly, residuals, objective contributions and solver status. The display must not show a single centered body as if it were the recovered geology. Independent prism checks and noise-covariance diagnostics have separate evidence from optimizer convergence. Implementation details are in data-pipeline/magnetic_forward.py, data-pipeline/magnetic_fit_assembly.py, and data-pipeline/magnetic_likelihood.py; source-bound tests include tests/data/test_magnetic_fit_assembly.py and tests/numerics/test_magnetic_forward.py.

4.1 Potential-field inverses

The gravity and magnetic experiments use SimPEG rectangular-prism sensitivity operators [1, 2], explicit survey geometry, uncertainty-weighted data residuals, and spatial regularization. The canonical potential-field setup uses a 10,752-cell mesh and 256 surface observations (docs/research/scientific-implementation.md). L2 and smoothed-L1 IRLS solutions are distinct estimators, not interchangeable render styles. Magnetic cases include scalar induced assumptions and vector magnetization so that remanence can be exposed as model misspecification. Choclo prism values [8] provide an independent physical check for sign and scale in specified comparisons. Sensitivity and conditional noise ensembles are not interpreted as calibrated geological posterior intervals.

The current result table includes a gravity deep-body example for which weighted WRMS is 1, while model correlation is only 0.283 and recovered centroid depth is approximately 570 m versus a synthetic truth near 282 m. This is not a paradox: the inverse fits the observations at the chosen noise scale while regularization and limited information admit a different model. The structure is unresolved despite the weighted data fit.

4.2 MT, ERT, and waveform processing

The MT response is calculated through the complex impedance recursion for isotropic layers; differentiable physics-guided inversion is a related approach [3]. EDI ingestion is a transfer-function workflow and preserves conventions, units, rotations, masks, and stated variances; EDI is not treated as raw time-series data. Multiple optimizers (bounded least squares, Adam, and a small per-sounding neural parameterization) address matched objectives in synthetic fixtures. A neural architecture related conceptually to physics-guided inversion [3] is not asserted to reproduce the cited paper's architecture or published results.

ERT workflows validate electrode/ABMN structure, QC and topographic factors before inversion, following established 2D methods such as PyGIMLi [9]. The numerical examples distinguish authored controls from absent or restricted field assets and retain failed inversions. Seismic preprocessing distinguishes instrument counts, response-corrected units, filters, gaps, and sample windows. STA/LTA picking and M13 learned inference consume explicitly specified traces; analyst picks are reference annotations, not subsurface truth.

4.3 Travel-time tomography, FWI, and joint inversion

Travel-time inversion uses specified ray geometry and a training/withheld partition. Its velocity maps are interpreted together with ray support; unsupported cells do not receive equivalent evidential status. The M09 bilingual lesson calls the product inverse core on an authored refracting two-layer survey and compares complete withheld shots with a training-only homogeneous baseline. This is a bounded synthetic calculation.

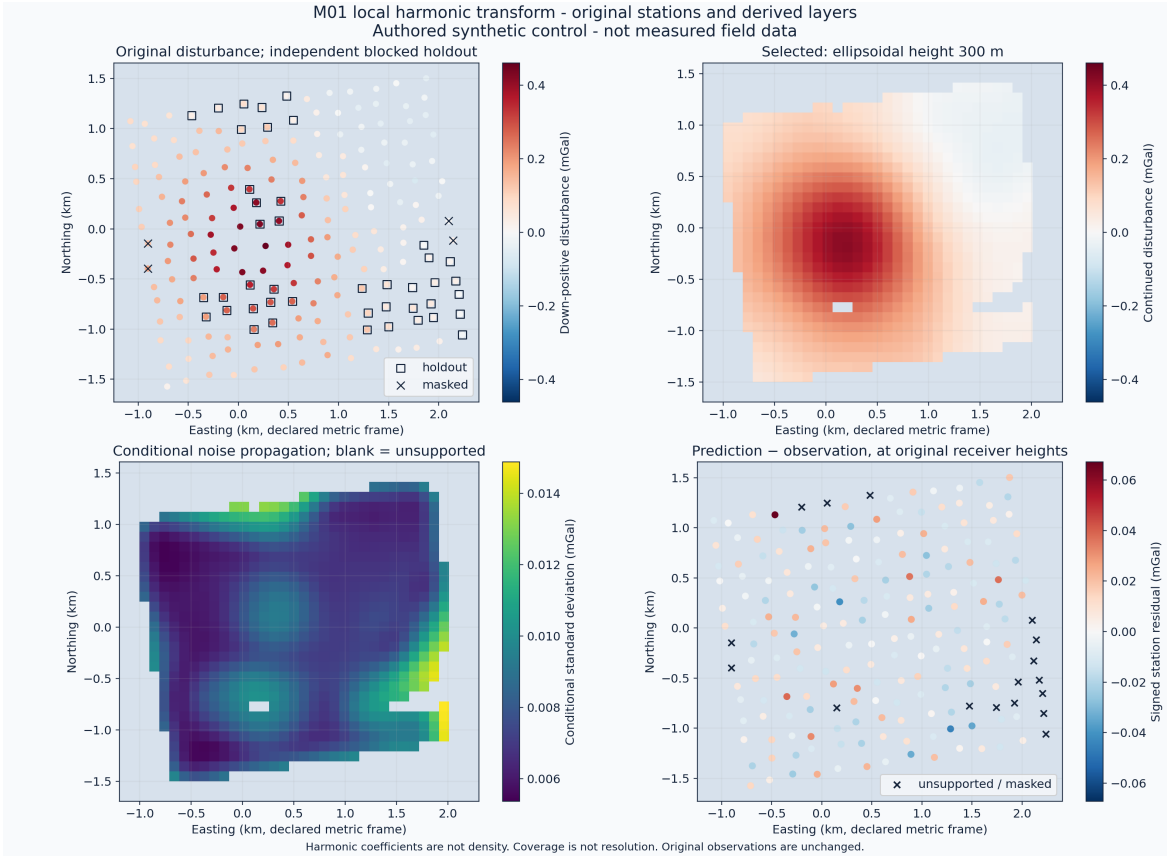


Figure 1: M01 authored synthetic control. Upper left: original station disturbance and blocked holdout or masked locations. Upper right: harmonic continuation to 300 m. Lower left: propagated conditional noise standard deviation, with blank cells unsupported. Lower right: signed prediction residuals at original receiver locations. The figure is synthetic and does not represent field measurements.

The FWI implementation uses constant-density acoustic propagation [4, 5], a fixed acquisition family, bounded velocity, a data-derived coarse background, and matched update budgets for full-band and frequency-continuation schedules. Active and withheld receiver WRMS accompany model RMSE. A lower model RMSE can coexist with residuals above the specified noise level. The four-case computation includes a layered reflection, a fault, a salt/cycle-skipping negative control, and a low-velocity channel. Its case-level results are not generalized to field-scale FWI or elastic/attenuating physics.

Joint inversion compares uncoupled and coupled estimates against the same observations and independent baselines. Cross-gradient coupling promotes structural similarity, while a Gaussian-mixture prior adds a distinct petrophysical assumption [10, 11]. These mechanisms are reported separately. In the shared synthetic case, cross-gradient slightly improves susceptibility RMSE while worsening density RMSE relative to the optimized uncoupled solution; the mixture-prior method slightly improves density RMSE while slightly worsening susceptibility. Neither is a joint two-property recovery gain. In a deliberately conflicting case, the negative control rejects a general claim that structural or petrophysical coupling must help.

4.4 Learned methods and residual refinement

The original gravity CNN predicts column density rather than a depth-resolved volume, reflecting the non-identifiability of depth from one potential-field map. Learned inverse benchmarks also show sensitivity to the sampled geological families [12]. Its comparison with a spatial L2 baseline uses identical seeded noisy observations and a shared column target. The autoencoder's threshold is selected on a separate calibration set and tested on withheld geometries; its measured failure to detect the withheld family is retained. These models are compact baselines and have not been compared with a broader set of published inverse methods.

The currently submitted BS-1 paper reports a geometry-normalized, coverage-conditioned residual CNN against a validation-selected weighted straight-ray reference. Its input has two channels (normalized residual and coverage); the network has 24 hidden feature maps, not 24 input channels. The fixed protocol uses seed 77213, 40 epochs, 800 training cases, 160 validation cases, and held-out identity, family, acquisition, and joint shifts. On the 160-case joint holdout, the primary checkpoint has velocity and independent travel-time RMSE ratios 2.194 and 5.585; the second CUDA execution, with the same seed and manifests, has ratios 2.190 and 5.595. All 160 joint cases exceed each run's validation-derived threshold. The paper therefore records a negative result for its specified generator and operators. Repeating the same seed measures execution sensitivity; it does not estimate between-seed training variance. The source-bound receipts, checkpoints, bootstrap calculation and current limitations are in manuscripts/geometry-normalized-traveltime-refinement-v001/.

An earlier residual-v3.1/null-space experiment and its analysis remain archived, but they are not the current BS-1 experiment. Their older cohort, checkpoint, scorer and operator evidence cannot be combined with the current results. The archive is retained for provenance at docs/research/archive/m12-nullspace-evaluation-draft-2026-11-13/; the product no longer declares it as a separate proposed-method paper. This distinction prevents the old positive aggregate ratios from being read as results of the current method or as a beyond-state-of-the-art claim.

5 Selected results and interpretation

5.1 Reference-condition outcomes

The reference-condition summary lists 122 recovered, 184 unresolved, 6 failed, and 36 negative-control method results (source: docs/validation/results.md). These are case-specific outcomes. They must not be summed into an overall accuracy or treated as success probabilities because the cases, units, target variables, and criteria differ. Failure and unresolved outcomes are first-class evidence in this project.

5.2 FWI case matrix

The bilingual case-matrix lesson reads four hash-verified full-run arrays (notebooks/en/06-seismology/05-m10-fwi-case-matrix.ipynb; Spanish counterpart). Layered reflection and fault are labeled recovered under the case-specific criteria; salt remains a cycle-skipping negative control; and the low-velocity channel remains unresolved. For the channel, model RMSE improves relative to the starting model (ratio 0.521 for the continuation result), but fitted/withheld WRMS are 2.071/2.213 against the specified unit noise scale.

This contrast shows why improvement from a starting model is not sufficient. The notebook re-analyzes saved output rather than re-solving FWI, and an independent propagator comparison is needed to test sensitivity to using the same forward engine.

5.3 EDI screening and phase picker

The observed cl061 EDI tensor is screened against necessary 1D consistency conditions and fails them. It remains QC-only. It provides no 1D resistivity model for cl061 and makes no geological interpretation from that station. For M13, the 2026-10-27 local browser parity computation reports all 23 valid selected display traces matching CPU ONNX Runtime: maximum absolute probability difference 5.0664×10^{-6}, no P/S peak displacement, and matching accepted/abstained picks within one sample. A twenty-fourth selected trace is QC-rejected and never inferred. The separate locked 6,000-trace benchmark, not this curated 23-trace subset, is the population-level performance record. Even that benchmark compares predictions with analyst annotations and should not be described as geological truth.

5.4 Novelty and null-space assessment

Null-space methods are established approaches for examining non-uniqueness in traveltime tomography and seismic inversion [17, 18, 19]. The archived product draft does not introduce null-space projection as new inverse theory. Its post-hoc projection experiment also does not provide a matched current checkpoint/operator comparison that would isolate projection as the cause of an improvement. The archived aggregate ratios and threshold diagnostics remain historical calculations for their stated frozen cohorts; they are not outcomes of the current BS-1 paper and do not establish superiority over published methods.

The current geometry-normalized residual study is likewise a bounded synthetic comparison, not a novel solver claim. It fails both pre-registered joint-shift error ratios for both same-seed CUDA runs. The scientifically supported contribution is therefore a reproducible negative comparison with explicit operators, cohort identities, per-case diagnostics and a disclosed replay discrepancy. A new method-performance claim would require an independent-seed training protocol, broader generators, matched contemporary baselines, exact operator controls and validation on eligible field observations. No such claim is made here.

6 Educational and visualization design

The bilingual notebook sequence covers inverse-problem foundations, gravity, magnetics, MT, ERT, seismology, joint inversion, and learning. Lessons include progressively deeper content: from units and forward physics to actual solver calls, held-out evaluation, sensitivity analyses, independent oracles, and negative controls. The notebooks are intended as research and teaching materials, but no educational efficacy study has been performed. The source course is cited as inspiration; its instructional materials are not copied into the product.

Scientific visualization is organized around the method's actual state variables. Potential-field scenes show geometry, cells, survey observations, cuts, and physical property scales; MT views show layer structure and frequency response; seismic instruments relate wavefields, receiver gathers, and inversion state; learned methods show their actual target

arrays and comparison metrics. Camera rotation changes viewpoint, not physics. Time cursors and optimizer-state replay address different quantities and are labeled accordingly. A rendered state alone is not evidence of uniqueness, resolution, or uncertainty.

The scientific views expose physical geometry, survey support, and computed states. Their effect on interpretation has not been measured in a user study, and browser performance has not been characterized across devices, routes, languages, and themes.

7 Computational reproducibility

All 60 bilingual notebooks were run from top to bottom in fresh kernels on 30 October 2026. The computation comprised 57 unchanged notebooks and three refreshed notebooks; none ended with an execution error. Focused fresh-kernel replays of the refreshed examples then passed. The complete replay used `scripts/run_notebooks.py --check` with external scratch storage. Notebook sources are under `notebooks/en/` and `notebooks/es/`.

Numerical outcomes are tied to their method-specific computation and date. The FWI case matrix reads hash-verified outputs and does not rerun the expensive inverse; its source is notebooks/en/06-seismology/05-m10-fwi-case-matrix.ipynb and the Spanish counterpart. The M12 notebook computation dated 28 October and browser-parity computation dated 27 October are documented computations in docs/validation/m12-checkpoint-notebook-2026-10-28.md and docs/validation/m13-browser-population-parity-2026-10-27.md. The null-space refinement results and cohort definitions are in docs/design/features/m12-velocity-validation/nullspace-v1_1-protocol.md. Reference-condition metrics are in docs/validation/results.md. Method equations, implementation details, and primary-source mapping are in docs/research/scientific-implementation.md, docs/research/review.md, and docs/design/SDD.md.

Notebook replay verifies execution for the recorded source and environment; it is not an independent physical oracle. Scientific replication requires using the method-specific data, environment, split, seed, operator, and comparison criteria together. A lesson that reads an archived computation must not be described as having recomputed that solve.

8 Limitations and open research questions

1. **Field validation and eligibility.** Most quantitative recovery results are synthetic. Provider data with incomplete datum, uncertainty, rights, or provenance remain unresolved or ineligible. cl061 is a QC failure for 1D inversion, not a field inverse.
2. **Non-uniqueness and uncertainty.** Conditional ensembles do not sample geological posterior uncertainty. Potential-field structure remains prior- and parameterization-dependent. Coverage and sensitivity do not by themselves yield calibrated resolution.
3. **FWI scope.** The experiments use bounded acoustic physics and fixed synthetic acquisition families. Source uncertainty, realistic correlated noise, anisotropy, elasticity, attenuation, broader cycle-skipping regimes, and an independent propagator remain necessary research directions.
4. **Joint inversion.** Cross-gradient and mixture-prior outcomes depend on their scale, prior, and structural compatibility. The limited calibration matrix does not establish general superiority over optimized single-modality estimates.
5. **Machine learning.** The M12 learned-velocity calculation fails its pre-specified joint-shift comparison. The positive null-space variant is narrow synthetic evidence and

needs external reproduction and broader cohorts. The phase-picker's 23-trace parity set is not an accuracy cohort; the separate 6,000-trace benchmark uses analyst annotations whose uncertainty limits the interpretation of timing scores.

6. **Data and reproducibility.** Some numerical tests require retained external inputs, including joint-inversion matrices and magnetic controls. Missing inputs are not replaced with synthetic stand-ins. Rights-limited source files are not copied merely to make a computation executable.
7. **Service capacity.** Local solver tests, builds, and browser parity do not establish worker admission, VPS capacity, cancellation, or recovery. Replacement-service capacity and operational behavior remain unmeasured.
8. **Educational effectiveness.** A bilingual executable notebook collection is a learning resource, but no controlled study establishes learning outcomes or comparative pedagogical effectiveness.

Useful next studies include preregistered geological-family and acquisition/noise ensembles; matched contemporary travel-time baselines; independent-seed training and calibration under forward-model and geological uncertainty; independent FWI propagation; licensed field inversions with source/datum/error review; and user studies of whether linking assumptions, predictions, residuals, and uncertainty improves interpretation.

9 Conclusions

Inverse Earth Studio combines source-linked geophysical processing and inverse methods with explicit assumptions, independent physical checks where available, and preserved negative evidence. Its central examples show that data fit, model recovery, transfer, and visual plausibility are different questions. The workbench includes synthetic and bounded real-data screening calculations, bilingual executable notebooks, and computational paths across multiple physical methods.

These computations do not establish field validity or generality across geological settings. Several method-specific comparisons, end-to-end user-supplied-data workflows, actual-host measurements, independent solver comparisons, and user-facing visual evaluations remain to be completed. The current learned travel-time refinement fails its two joint-shift error criteria in the primary run and a same-seed replay. The earlier null-space draft is archived as a historical analysis of an established operation, not as a novel method. The work provides a reproducible software and experimental framework that makes assumptions and counterexamples inspectable. Algorithmic novelty and publication priority are not established.

10 Data and code availability

Source code is available at `https://github.com/fsantibanezleal/CAOS_Geophysics`. Scientific methods, notebooks, test data, computational records, and source-rights decisions are linked above and organized in `data/`, `data-pipeline/`, `docs/`, `notebooks/`, `tests/`, and `models/`. Some external raw observations are absent because redistribution rights or field eligibility remain unresolved. The intended application origin is `https://geophysics.ml.fasl-work.com/`. Local computations do not establish replacement-service capacity.

11 Acknowledgements

The work uses and cites open scientific software and public geophysical data sources listed below and in the repository's research review. No endorsement by the cited software or dataset authors is implied.

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